UNIVERSITY of RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF IFORMATION SYSTEMS

END OF SEMESTER EXAMINATION -ACADEMIC YEAR 2025-2026

YEAR: 2 **SEMESTER: I** **GROUPS: IS, CSE, CS**

MODULE CODE & TITLE: IS80361 – COMPUTER NETWORKS

DATE: 23/01/2026

TIME: 2hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains **FOUR (4)** questions.
2. **Answer THREE (3) Questions only:**
   Question ONE (1) from Section "A" is Compulsory and Answer any TWO (2) from Section "B"
3. No written materials allowed.
4. Do not forget to write your Registration Number.
5. Write all your answers in the booklet provided
6. Do not write any answer on this questions paper.
7. Start each question in a NEW page

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SECTION A

Compulsory

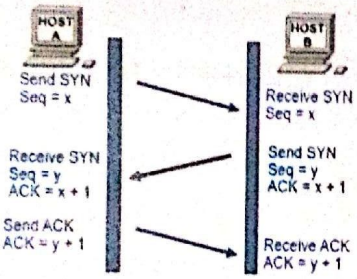
Question 1 **(20 marks)**

a) Define a protocol and explain three key elements of a protocol. **(2pts)**

b) Using the LRC what will be the redundant bit pattern if an even parity check is used for the following data? **(2pts)**

c) What is the Network IP address does the host IP address: **172.20.243.138/27** belongs in? **(2pts)**

d) In the following figure, how is process called? Which protocol perform this process? **(2pts)**



e) Which OSI model layers are implemented in the host? **(2 pts)**

f) What is the Subnet Mask corresponding to a shorthand of /11? **(2 pts)**

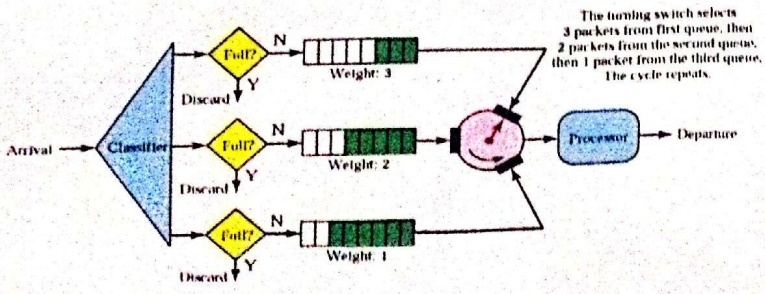
g) **Reliability** and **Scalability** are some of the criteria in designing a network. Explain both criteria. **(2 pts)**

h) Write the following IPv6 address by using the short notation:
00DF:0000:BB00:0004:FACE:0000:0000:0000 **(2 pts)**

i) Provide four main functions of data link layer. **(2 pts)**

j) Give two examples of interior gateway protocols **(1 pt)**

k) Which type of queuing in the figure below



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SECTION B

Attempt any 2 questions

Question 2

(15 marks)

a) Suppose you are hired as a network administrator and your task is to configure the following network using the an IP address: 192.168.10.192/26. Complete the following addressing table, given that each router interface is assigned the first usable address. Provide the First usable IP addresses to all Gigabit Ethernet interfaces, S0/0/0 of RT-X, and S0/0/1 of RT-Y. The remaining serial interfaces will be given the second (last) usable addresses. Web & DHCP Server should have the last usable IP address of its subnet.

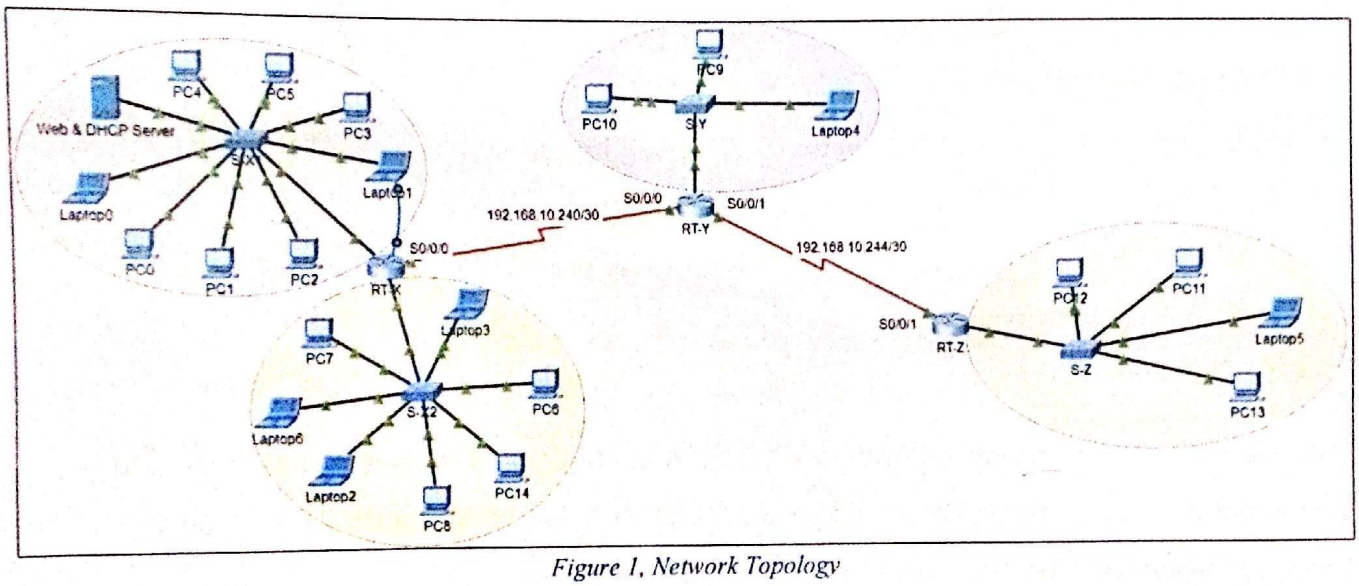


Figure 1, Network Topology

IPv4 addressing table

Device	Interface	IP address	Subnet mask	Default Gateway
RT-X	G0/0 (biggest LAN) First			
	G0/1 First			
	S0/0/0 First			
RT-Y	G0/0 First			
	S0/0/0 Last			
	S0/0/1 First			
RT-Z	G0/0 Last			
LAN3	S0/0/1 Last			
Web & DHCP Server	NIC Last			

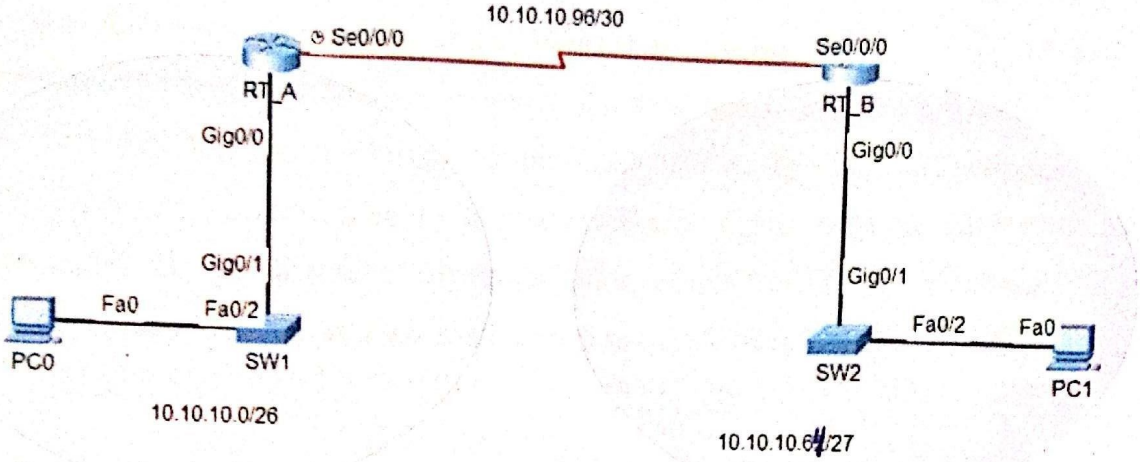
b) List the remaining subnet IP addresses (with their prefix length) **(4 pts)**

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Question 3

(15 marks)

Consider the network scenario below:



a) Specify each subnet with its network ID (IP address). **(3 marks)**

b) Configure IP addresses on each router's connected interface **(6 pts)**

c) Configure OSPF routing protocol on RT-B. Use process-id = 10 and save all the configurations **(6 pts)**

Question 4

(15 marks)

You are hired by a company to design and install a single Local Area Network (LAN) with a maximum of 10 PCs. The LAN should be connected on Internet. Using the TCP/IP reference model, indicate the activities performed at each layer.

a) Sketch a network topology of your choice. **(3 pts)**

b) Using the table provided, indicate the tasks to be performed step by step, layer by layer, according to the TCP/IP model. **(12 pts)**

Layer Name	Activities
Ap	
Trq	
Ntw	
Nif	

Good luck!!!

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